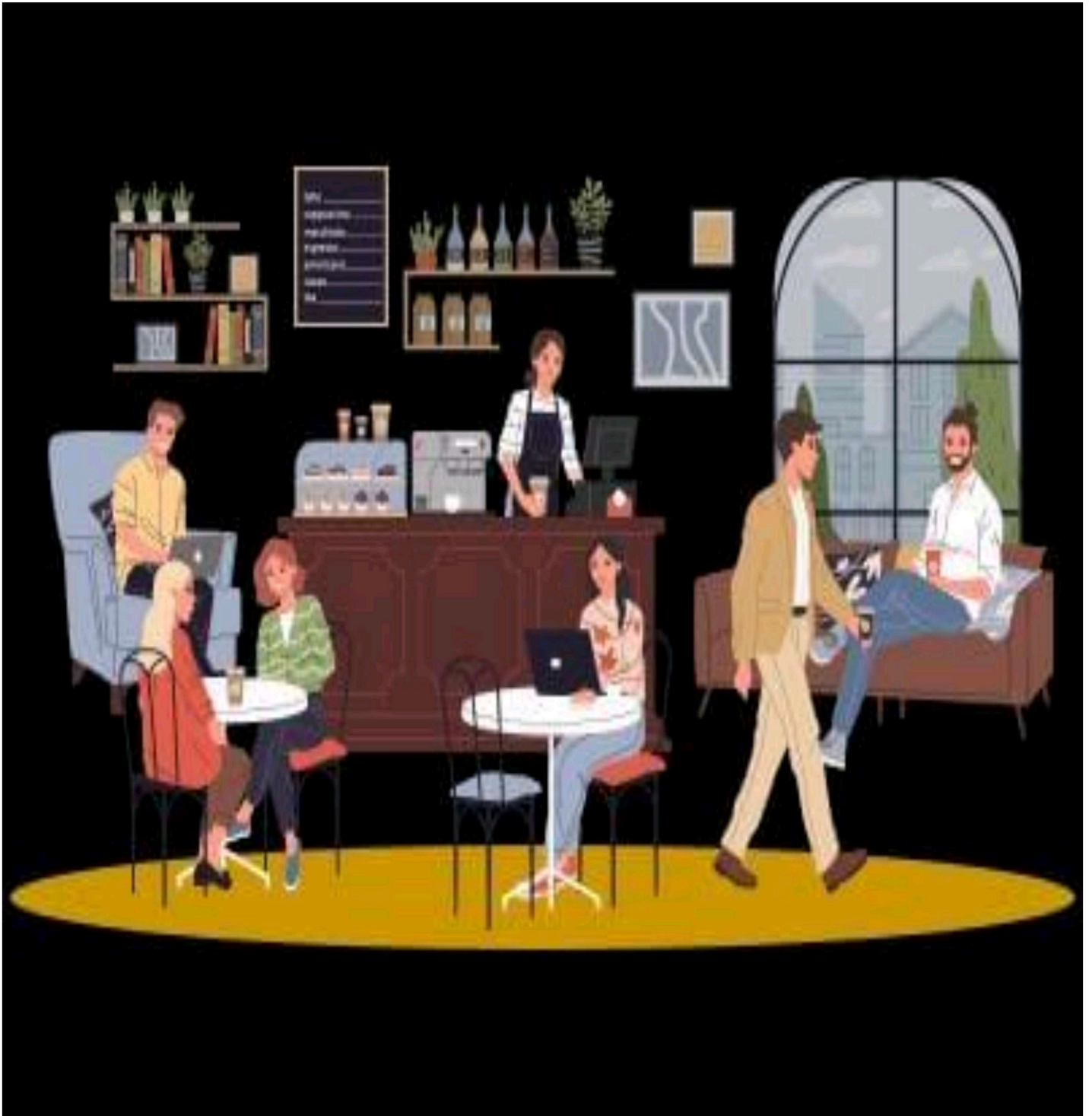


Coffee Shop Management System



Coffee Shop Management System :-

The Coffee Shop Management System is a relational database designed to manage daily cafe operations efficiently. The system handles customer orders, menu items, recipes, ingredient, inventory, staff details, and staff shift scheduling. The database is designed using SQL with proper normalization and relational integrity. This project includes entities and attributes related to coffee shop management system and relation between them at appropriate place. All functionalities and knowledge of SQL that I learned are performed in this project and all the queries are executed successfully.

Objectives :-

The objectives of this project are :-

- To design a normalized relational database for a coffee shop
- To manage transactional and master data efficiently
- To enforce data integrity using primary and foreign keys
- To support analytical queries for business insights

Database Design Overview:-

The database consists of **8 core entities**:

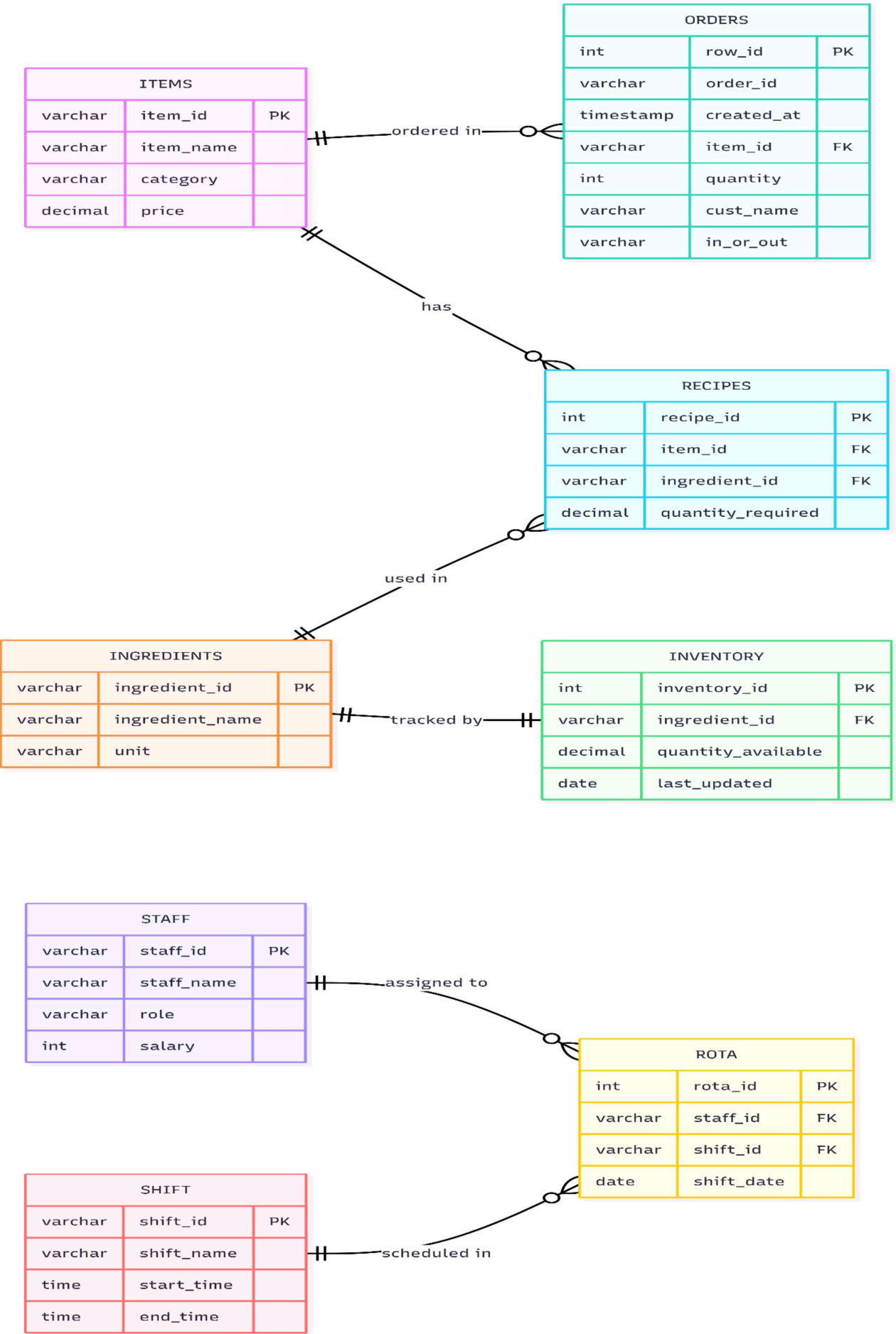
- Items
- Orders
- Recipes
- Ingredients
- Inventory
- Staff
- Shift
- Rota

The design follows **Third Normal Form (3NF)** to eliminate redundancy and improve scalability.

Tools & Technologies Used :-

- SQL (MySQL)
- Relational Database Design
- ER Modelling
- Constraints & Foreign Keys

ER Diagram Coffee Shop Management System



Learning Outcome :-

Through this project, I strengthened my skills in:

- Database design & normalization
- Writing optimized SQL queries
- Understanding real-world business requirements
- Translating requirements into relational models

Future Enhancements :-

- Normalize orders into orders & order_details
- Add triggers for inventory auto-update
- Create views for sales and inventory reports
- Build Power BI / Tableau dashboards

Conclusion :-

This project successfully demonstrates the design and implementation of a real-world coffee shop management database.

The system supports transactional processing, inventory tracking, and staff scheduling while maintaining data integrity and scalability.

[Open Drive Folder for SQL Code :-](#)

<https://drive.google.com/drive/folders/1OZAUtAyiDijlOllvn98MBsQ-5kGGLUpM?usp=sharing>

SQL Script

/* 1:

Creating new database with the name CoffeeShopManagementDB.*/

create database CoffeeShopManagementDB;

use CoffeeShopManagementDB;

/* 2:

Creating new table with name items consisting of item_id,item_name,category & price.*/

```
create table items (
    item_id VARCHAR(10) PRIMARY KEY,
    item_name VARCHAR(100),
    category VARCHAR(50),
    price DECIMAL(6,2)
);
```

/* 3:

Insert data into table items.*/

insert into items (item_id,item_name,category,price) values

```
('It001','Espresso','Coffee',120),
('It002','Double Espresso','Coffee',150),
('It003','Cappuccino','Coffee',150),
('It004','Caffe Latte','Coffee',160),
('It005','Vanilla Latte','Coffee',170),
('It006','Caramel Latte','Coffee',180),
('It007','Hazelnut Latte','Coffee',180),
('It008','Americano','Coffee',140),
('It009','Long Black','Coffee',145),
('It010','Mocha','Coffee',170),
('It011','Flat White','Coffee',160),
('It012','Macchiato','Coffee',155),
('It013','Affogato','Coffee',190),
('It014','Cold Coffee','Cold Drink',180),
('It015','Iced Latte','Cold Drink',190),
('It016','Lemon Iced Tea','Cold Drink',140),
('It017','Peach Iced Tea','Cold Drink',150),
('It018','Green Tea','Tea',110),
('It019','Masala Tea','Tea',100),
('It020','Black Tea','Tea',90),
('It021','Hot Chocolate','Beverage',200),
('It022','White Hot Chocolate','Beverage',210),
('It023','Chocolate Brownie','Bakery',120),
('It024','Chocolate Muffin','Bakery',90),
('It025','Blueberry Muffin','Bakery',100),
('It026','Croissant','Bakery',110),
('It027','Cheese Sandwich','Snack',130);
```

```
/* 4:  
Creating new table with name ingredients consisting of ingredient_id,ingredient_name &  
unit.*/
```

```
create table ingredients (  
    ingredient_id VARCHAR(10) PRIMARY KEY,  
    ingredient_name VARCHAR(100),  
    unit VARCHAR(20)  
);
```

```
/* 5:  
Insert data into table ingredients.*/  
insert into ingredients (ingredient_id,ingredient_name,unit) values  
( 'Ing01','Coffee Beans','grams'),  
( 'Ing02','Milk','ml'),  
( 'Ing03','Sugar','grams'),  
( 'Ing04','Cocoa Powder','grams'),  
( 'Ing05','Tea Leaves','grams'),  
( 'Ing06','Chocolate Syrup','ml'),  
( 'Ing07','Flour','grams'),  
( 'Ing08','Butter','grams'),  
( 'Ing09','Yeast','grams'),  
( 'Ing10','Salt','grams'),  
( 'Ing11','Vanilla Syrup','ml'),  
( 'Ing12','Caramel Syrup','ml'),  
( 'Ing13','Hazelnut Syrup','ml'),  
( 'Ing14','Ice Cubes','units'),  
( 'Ing15','Lemon','units'),  
( 'Ing16','Peach Syrup','ml'),  
( 'Ing17','Water','ml'),  
( 'Ing18','Cream','ml'),  
( 'Ing19','Cheese','grams'),  
( 'Ing20','Chocolate Chips','grams');
```

```
/* 6:  
Creating new table with name recipes consisting of recipe_id,item_id,ingredient_id &  
quantity_required.*/
```

```
create table recipes (  
    recipe_id INT PRIMARY KEY,  
    item_id VARCHAR(10),  
    ingredient_id VARCHAR(10),  
    quantity_required DECIMAL(6,2),  
    FOREIGN KEY (item_id) REFERENCES items(item_id),  
    FOREIGN KEY (ingredient_id) REFERENCES ingredients(ingredient_id)  
);
```

```

/* 7:
Insert data into table recipes.*/
insert into recipes (recipe_id,item_id,ingredient_id,quantity_required) values
(1,'It001','Ing01',18),
(2,'It002','Ing01',22),
(3,'It003','Ing01',15),
(4,'It003','Ing02',120),
(5,'It004','Ing02',150),
(6,'It005','Ing11',20),
(7,'It006','Ing12',20),
(8,'It007','Ing13',20),
(9,'It008','Ing01',18),
(10,'It009','Ing01',18),
(11,'It010','Ing04',10),
(12,'It011','Ing02',100),
(13,'It012','Ing01',14),
(14,'It013','Ing18',30),
(15,'It014','Ing14',6),
(16,'It015','Ing02',160),
(17,'It016','Ing15',1),
(18,'It017','Ing16',25),
(19,'It018','Ing05',6),
(20,'It019','Ing05',8),
(21,'It020','Ing05',7),
(22,'It021','Ing06',25),
(23,'It022','Ing06',30),
(24,'It024','Ing07',100),
(25,'It027','Ing19',60);

```

```

/* 8:
Creating new table with name inventory consisting of
inventory_id,ingredient_id,quantity_available & last_updated.*/
create table inventory (
    inventory_id INT PRIMARY KEY,
    ingredient_id VARCHAR(10),
    quantity_available DECIMAL(8,2),
    last_updated DATE,
    FOREIGN KEY (ingredient_id) REFERENCES ingredients(ingredient_id)
);

```

```

/* 9:
Insert data into table inventory.*/
insert into inventory (inventory_id,ingredient_id,quantity_available,last_updated)
values
(1,'Ing01',6000,'2024-02-12'),
(2,'Ing02',12000,'2024-02-12'),
(3,'Ing03',4000,'2024-02-12'),
(4,'Ing04',2000,'2024-02-12'),
(5,'Ing05',1800,'2024-02-12'),
(6,'Ing06',2500,'2024-02-12'),
(7,'Ing07',5000,'2024-02-12'),
(8,'Ing08',3000,'2024-02-12'),
(9,'Ing09',1500,'2024-02-12'),
(10,'Ing10',1000,'2024-02-12'),
(11,'Ing11',1800,'2024-02-12'),
(12,'Ing12',1700,'2024-02-12'),

```



```
(13, 'Ing13', 1600, '2024-02-12'),
(14, 'Ing14', 500, '2024-02-12'),
(15, 'Ing15', 300, '2024-02-12'),
(16, 'Ing16', 900, '2024-02-12'),
(17, 'Ing17', 20000, '2024-02-12'),
(18, 'Ing18', 2500, '2024-02-12'),
(19, 'Ing19', 2200, '2024-02-12'),
(20, 'Ing20', 1800, '2024-02-12');
```

/* 10:

Creating new table with name orders consisting of
row_id, order_id, created_at, item_id, quantity, cust_name & in_or_out.*/

```
create table orders (
    row_id INT PRIMARY KEY,
    order_id VARCHAR(10) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    item_id VARCHAR(10) NOT NULL,
    quantity INT NOT NULL CHECK (quantity > 0),
    cust_name VARCHAR(100) NOT NULL,
    in_or_out VARCHAR(10) NOT NULL CHECK (in_or_out IN ('in', 'out'))
);
```

/* 11:

Insert data into table orders.*/

```
insert into orders (row_id, order_id, created_at, item_id, quantity, cust_name, in_or_out)
values
```

```
(1, 'ORD001', '2024-02-12 07:04:00', 'It008', 1, 'Alex', 'out'),
(2, 'ORD002', '2024-02-12 07:09:00', 'It014', 1, 'Jordan', 'in'),
(3, 'ORD003', '2024-02-12 07:14:00', 'It003', 2, 'Taylor', 'out'),
(4, 'ORD004', '2024-02-12 07:18:00', 'It019', 1, 'Casey', 'out'),
(5, 'ORD005', '2024-02-12 07:23:00', 'It024', 2, 'Jamie', 'out'),
(6, 'ORD006', '2024-02-12 07:27:00', 'It005', 1, 'Morgan', 'in'),
(7, 'ORD007', '2024-02-12 07:32:00', 'It010', 1, 'Riley', 'out'),
(8, 'ORD008', '2024-02-12 07:36:00', 'It022', 1, 'Drew', 'out'),
(9, 'ORD009', '2024-02-12 07:41:00', 'It011', 2, 'Quinn', 'out'),
(10, 'ORD010', '2024-02-12 07:45:00', 'It001', 1, 'Avery', 'in'),
(11, 'ORD011', '2024-02-12 07:50:00', 'It026', 1, 'Parker', 'out'),
(12, 'ORD012', '2024-02-12 07:54:00', 'It018', 1, 'Blake', 'out'),
(13, 'ORD013', '2024-02-12 07:59:00', 'It015', 2, 'Cameron', 'out'),
(14, 'ORD014', '2024-02-12 08:03:00', 'It002', 1, 'Dakota', 'in'),
(15, 'ORD015', '2024-02-12 08:08:00', 'It004', 1, 'Reese', 'out'),
(16, 'ORD016', '2024-02-12 08:12:00', 'It007', 1, 'Skyler', 'out'),
(17, 'ORD017', '2024-02-12 08:16:00', 'It017', 2, 'Rowan', 'in'),
(18, 'ORD018', '2024-02-12 08:21:00', 'It021', 1, 'Emerson', 'out'),
(19, 'ORD019', '2024-02-12 08:25:00', 'It023', 1, 'Finley', 'out'),
(20, 'ORD020', '2024-02-12 08:30:00', 'It006', 2, 'Harper', 'in'),
(21, 'ORD021', '2024-02-12 08:34:00', 'It020', 1, 'Sawyer', 'out'),
(22, 'ORD022', '2024-02-12 08:39:00', 'It009', 1, 'Logan', 'out'),
(23, 'ORD023', '2024-02-12 08:43:00', 'It012', 1, 'Ellis', 'in'),
(24, 'ORD024', '2024-02-12 08:48:00', 'It025', 2, 'Phoenix', 'out'),
(25, 'ORD025', '2024-02-12 08:52:00', 'It027', 1, 'River', 'out');
```

```
/* 12:
Creating new table with name staff consisting of staff_id,staff_name,role & salary.*/
create table staff (
    staff_id VARCHAR(10) PRIMARY KEY,
    staff_name VARCHAR(100),
    role VARCHAR(50),
    salary INT
);
```

```
/* 13:
Insert data into table staff.*/
insert into staff (staff_id,staff_name,role,salary) values
('S01','Aman','Barista',18000),
('S02','Riya','Cashier',16000),
('S03','Karan','Barista',18000),
('S04','Neha','Manager',30000),
('S05','Rohit','Cleaner',12000),
('S06','Ankit','Barista',17500),
('S07','Pooja','Cashier',16500),
('S08','Rahul','Supervisor',22000),
('S09','Sneha','Barista',18000),
('S10','Vikas','Cleaner',12000),
('S11','Arjun','Barista',18500),
('S12','Nidhi','Cashier',16000),
('S13','Manish','Supervisor',22500),
('S14','Kriti','Barista',17800),
('S15','Suresh','Cleaner',11500),
('S16','Meena','Cashier',15800),
('S17','Deepak','Barista',18200),
('S18','Isha','Barista',17600),
('S19','Yash','Cleaner',11800),
('S20','Anjali','Manager',31000);
```

```
/* 14:
Creating new table with name shift consisting of shift_id,shift_name,start_time &
end_time.*/
create table shift (
    shift_id VARCHAR(10) PRIMARY KEY,
    shift_name VARCHAR(50),
    start_time TIME,
    end_time TIME
);
```

```
/* 15:
Insert data into table shift.*/
insert into shift (shift_id,shift_name,start_time,end_time) values
('SH01','Morning','06:00','12:00'),
('SH02','Afternoon','12:00','18:00'),
('SH03','Evening','18:00','23:00'),
('SH04','Night','23:00','06:00'),
('SH05','Weekend Morning','07:00','13:00'),
('SH06','Weekend Evening','14:00','22:00'),
('SH07','Early Morning','05:00','11:00'),
('SH08','Late Afternoon','15:00','21:00'),
```

```
('SH09','Split Shift A','08:00','12:00'),
('SH10','Split Shift B','16:00','20:00'),
('SH11','Holiday Morning','07:00','13:00'),
('SH12','Holiday Evening','14:00','22:00'),
('SH13','Festival Morning','06:00','12:00'),
('SH14','Festival Evening','16:00','22:00'),
('SH15','Backup Shift','10:00','16:00'),
('SH16','Overtime Morning','06:00','10:00'),
('SH17','Overtime Evening','18:00','22:00'),
('SH18','Training Shift','11:00','15:00'),
('SH19','Maintenance Shift','22:00','02:00'),
('SH20','Inventory Shift','04:00','08:00');
```

/* 16:

Creating new table with name rota consisting of rota_id,staff_id,shift_id & shift_date.*/

```
create table rota (
    rota_id INT PRIMARY KEY,
    staff_id VARCHAR(10),
    shift_id VARCHAR(10),
    shift_date DATE,
    FOREIGN KEY (staff_id) REFERENCES staff(staff_id),
    FOREIGN KEY (shift_id) REFERENCES shift(shift_id)
);
```

/* 17:

Insert data into table rota.*/

```
insert into rota (rota_id,staff_id,shift_id,shift_date) values
(1,'S01','SH01','2024-02-12'),
(2,'S02','SH01','2024-02-12'),
(3,'S03','SH02','2024-02-12'),
(4,'S04','SH02','2024-02-12'),
(5,'S05','SH03','2024-02-12'),
(6,'S06','SH03','2024-02-12'),
(7,'S07','SH04','2024-02-12'),
(8,'S08','SH05','2024-02-12'),
(9,'S09','SH06','2024-02-12'),
(10,'S10','SH07','2024-02-12'),
(11,'S11','SH08','2024-02-12'),
(12,'S12','SH09','2024-02-12'),
(13,'S13','SH10','2024-02-12'),
(14,'S14','SH11','2024-02-12'),
(15,'S15','SH12','2024-02-12'),
(16,'S16','SH13','2024-02-12'),
(17,'S17','SH14','2024-02-12'),
(18,'S18','SH15','2024-02-12'),
(19,'S19','SH16','2024-02-12'),
(20,'S20','SH17','2024-02-12'),
(21,'S01','SH18','2024-02-13'),
(22,'S02','SH19','2024-02-13'),
(23,'S03','SH20','2024-02-13'),
(24,'S04','SH01','2024-02-13'),
(25,'S05','SH02','2024-02-13');
```


SQL Query Handling

/*Problem # 1:

Write an SQL query to find the total quantity sold for each item.*/

```
select item_id , sum(quantity) as total_quantity_sold
from orders
group by item_id;
```

/*Problem # 2:

Write an SQL query to display customers who ordered more than one quantity in a single order.*/

```
select cust_name , order_id, quantity
from orders
where quantity > 1;
```

/*Problem # 3:

Write an SQL query to calculate the total sales amount generated by each item.*/

```
select o.item_id , i.item_name , sum(o.quantity * i.price) as total_sales
from orders as o
join items as i
on o.item_id = i.item_id
group by o.item_id, i.item_name;
```

/*Problem # 4:

Write an SQL query to list all ingredients used in each menu item.*/

```
select i.item_name , g.ingredient_name
from recipes as r
join items as i
on r.item_id = i.item_id
join ingredients as g
on r.ingredient_id = g.ingredient_id;
```

/*Problem # 5:

Write an SQL query to display staff members working on a specific date.*/

```
select s.staff_name , sh.shift_name , r.shift_date
from rota as r
join staff as s
on r.staff_id = s.staff_id
join shift as sh
on r.shift_id = sh.shift_id
where r.shift_date = '2024-02-12';
```

/*Problem- # 6:

Write an SQL query to find ingredients that are not used in any recipe.*/

```
select ingredient_id , ingredient_name
from ingredients
where ingredient_id not in (select distinct ingredient_id from recipes);
```

/*Problem- # 7:

Write an SQL query to find the most expensive item in each category.*/

```
select category , item_name , price
from items as i
where price = (select max(price) from items where category = i.category);
```


/*Problem- # 8:

Write an SQL query to display hour-wise order count.*/

```
select hour(created_at) as order_hour , count(*) as total_orders
from orders
group by hour(created_at)
order by order_hour;
```

/*Problem- # 9:

Write an SQL query to rank items based on total sales amount.*/

```
select i.item_name , sum(o.quantity * i.price) as total_sales ,
rank() over (order by sum(o.quantity * i.price) DESC) as sales_rank
from orders as o
join items as i
on o.item_id = i.item_id
group by i.item_name;
```

/*Problem- # 10:

Write an SQL query to find staff members who worked more than one shift on the same day.*/

```
select staff_id , shift_date , count(*) as shift_count
from rota
group by staff_id , shift_date
having count(*) > 1;
```

/*Problem- # 11:

Write an SQL query to calculate total ingredient consumption based on all orders.*/

```
select i.ingredient_name , sum(o.quantity * r.quantity_required) as total_consumed
from orders as o
join recipes as r
on o.item_id = r.item_id
join ingredients as i
on r.ingredient_id = i.ingredient_id
group by i.ingredient_name;
```

/*Problem- # 12:

Write an SQL query to find ingredients where required quantity exceeds available inventory.*/

```
select g.ingredient_name , i.quantity_available , sum(o.quantity * r.quantity_required)
as total_required
from orders as o
join recipes as r
on o.item_id = r.item_id
join inventory as i
on r.ingredient_id = i.ingredient_id
join ingredients as g
on i.ingredient_id = g.ingredient_id
group by g.ingredient_name , i.quantity_available
having total_required > i.quantity_available;
```

/*Problem- # 13:

Write an SQL query to find the top 3 best-selling items based on quantity sold.*/

```
with item_sales as (  
  select i.item_name , sum(o.quantity) as total_quantity  
  from orders as o  
  join items as i  
  on o.item_id = i.item_id  
  group by i.item_name)  
select item_name , total_quantity  
from item_sales  
order by total_quantity DESC  
LIMIT 3;
```

/*Problem- # 14:

Write an SQL query to create a view that shows daily revenue.*/

```
create view daily_revenue as  
select date(o.created_at) as sale_date , sum(o.quantity * i.price) as total_revenue  
from orders as o  
join items as i  
on o.item_id = i.item_id  
group by date(o.created_at);
```

/*Problem- # 15:

Write a CTE (Common Table Expression) to rank items based on total sales.*/

```
with ranked_items as (  
  select i.item_name , sum(o.quantity * i.price) as total_sales,  
  rank() over (order by sum(o.quantity * i.price) DESC) as sales_rank  
  from orders as o  
  join items as i  
  on o.item_id = i.item_id  
  group by i.item_name  
)  
select * from ranked_items;
```

*All the SQL query executed successfully.